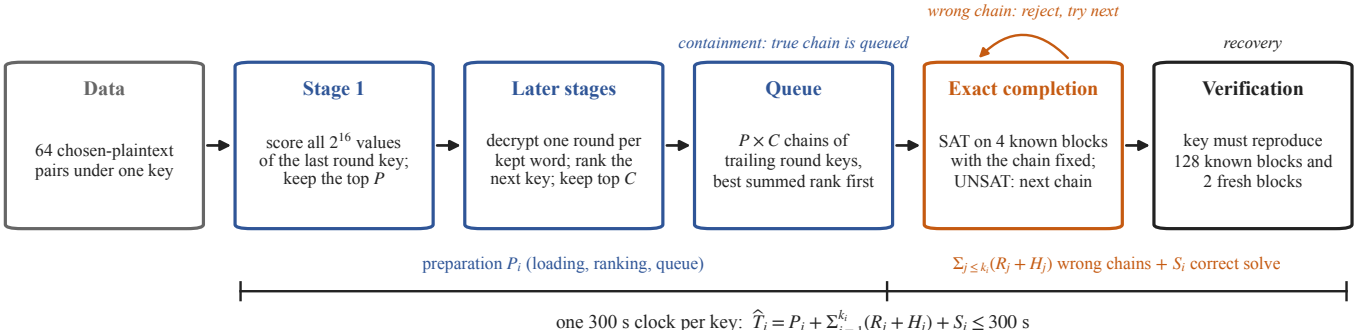

wrong chain: reject, try next
containment: true chain is queued
recovery
Data
64 chosen-plaintext pairs under one key
Stage 1
score all 2^{16} values of the last round key; keep the top P
Later stages
decrypt one round per kept word; rank the next key; keep top C
Queue
$P \times C$ chains of trailing round keys, best summed rank first
Exact completion
SAT on 4 known blocks with the chain fixed; UNSAT: next chain
Verification
key must reproduce 128 known blocks and 2 fresh blocks
preparation P_i (loading, ranking, queue)
$\Sigma_{j \le k_i}(R_j + H_j)$ wrong chains + S_i correct solve
one 300 s clock per key: $\hat{T}_i = P_i + \Sigma_{j=1}^{k_i}(R_j + H_j) + S_i \le 300$ s